

Q25 Final Conclusion:

Yes, the synthesized implementation preserves the functional behavior of the RTL for this testbench. Both RTL and GLS simulations show detected=1 at identical clock cycles for input sequence 1011, confirming logical correctness with only expected 2-3ns gate delay in GLS vs ideal zero-delay in RTL.

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https://fluffy-cod-wv769494g4wrf5qxw.github.dev

vsd-rtl [Codespaces: fluffy.cod]

Explorer

- vsd-rtl
- sky130RTLDesignAndSynthesisWorkshop
 - verilog_files
 - multiple_module...
 - multiple_module...
 - multiple_modules...
 - multiple_modules...
 - multiple_modules.v
 - mux_generate.v
 - mux_spice.v
 - net.v
 - opt_check.v
 - opt_check2.v
 - opt_check3.v
 - opt_check4.v
 - partial_case_assig...
 - pattern_detect_fs...
 - pattern_detect_fs...
 - rca.v
 - ripple_counter.v
 - schematic.q21**
 - seq_detector....
 - seq1011.dot
 - tb_bad_case.v
 - tb_bad_counter.v
 - th had latch.v

Outline

Timeline

Problems Output Debug Console Terminal Ports

bash - verilog_files

```
CPU: user 0.08s system 0.01s, MEM: 16.64 MB total, 11.23 MB resident
Yosys 0.9 (git sha1 1979e0b)
Time spent: 22% 16x opt_clean (0 sec), 17% 14x opt_expr (0 sec), ...
vscode@codespaces-79191d:/workspaces
/vsd-rtl/sky130RTLDesignAndSynthesisWorkshop/verilog_files$ dot -Tpng seq1011.dot -o schematic.q21.png
ls -l *.png
-rw-rw-rw- 1 vscode vscode 299714 Sep  7 06:27 schematic.q21.png
vscode@codespaces-79191d:/workspaces/vsd-rtl/sky130RTLDesignAndSynthesisWorkshop/verilog_files$ dot -Tpng seq1011.dot -o schematic.q21.png
ls -l *.pngschematic.q21.pngschematic.q21.pngschematic.q21.pngschematic.q21.pngschematic.q21.png
vscode@codespaces-79191d:/workspaces/vsd-rtl/sky130RTLDesignAndSynthesisWorkshop/verilog_files$
```

Build with Agent

AI responses may be inaccurate

Generate Agent Instructions to onboard AI onto your codebase.

Describe what to build

+ Agent Auto

Local Default permissions

Whole Image 1766x1416 Sign in 292.59KB Layout: US

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assessments/25: X Sign in - Google: X Anurag Assess: X vsdip/vsd-rtl: X assessments/25: X Anurag Assess: X assessments/25: X Anurag Assess: X [Preview] READ: X

https://fluffy-cod-wv769494g4wrf5qxw.github.dev

vsd-rtl [Codespaces: fluffy cod]

Yosys 0.9 (git sha1 1979e0b)

```
yosys> exit
End of script. Logfile hash: da39a3ee5e
CPU: user 0.00s system 0.01s, MEM: 15.69 MB total, 9.38 MB resident
Yosys 0.9 (git sha1 1979e0b)
Time spent: no commands executed
vscode@codespaces-bfa4d3:/workspaces/vsd-rtl$ iverilog
iverilog: no source files.

Usage: iverilog [-EiSuvV] [-B base] [-c cmdfile|-f cmdfile]
          [-g1995|-g2001|-g2005|-g2005-sv|-g2009|-g2012] [-g<feature>]
          [-D macro[=defn]] [-I includedir] [-L moduledir]
          [-M [mode=]depfile] [-m module]
          [-N file] [-o filename] [-p flag=value]
          [-s topmodule] [-t target] [-T min|typ|max]
```

Problems Output Debug Console Terminal Ports

```
vscode@codespaces-79191d:/workspaces/vsd-rtl/sky130RTLDesignAndSynthesis
Workshop/verilog_files$ 00m$ yosys -p "read verilog seq.detector.1011.v; synth; stat" 2>&1 | grep DFF
2.20.6. Executing OPT_RMDFF pass (remove dff with constant values).
2.20.13. Executing OPT_RMDFF pass (remove dff with constant values).
2.22.3. Executing OPT_RMDFF pass (remove dff with constant values).
2.24.3. Executing OPT_RMDFF pass (remove dff with constant values).
$ DFF_P_ 5
$ DFF_P_ 5
vscode@codespaces-79191d:/workspaces/vsd-rtl/sky130RTLDesignAndSynthesisworkshop/verilog_files$
```

Codespaces: fluffy cod main

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